A Dual Strategy Combining Spectral Shift Screening and Coarse-Grained Molecular Dynamics for $A_{2A}R$

Magdalena Szczuka[2,3], Vanessa Porkolab[1], Maud Sigoillot[1], Carsten Brock[1], Carleton Sage[4], Olivier Mirguet[4], Kerstin Kanonenberg[2], Matthieu Chavent[3]

1. Eurofins Cerep | France, 2. Eurofins CALIXAR | France, 3. LMGM CNRS | France 4. Eurofins Discovery Drug Discovery Chemistry | USA

The Adenosine A_{2A} Receptor ($A_{2A}R$) was selected as a representative GPCR to establish an integrated fragment-based discovery workflow capable of screening fragments on the active receptor state. This workflow combines Spectral Shift screening, coarse-grained Martini 3 molecular dynamics (MD) simulations and computational chemistry to efficiently identify and validate fragment hits. Using this approach, we discovered several fragment binders that were subsequently confirmed using thermal shift and cellular assays, providing mechanistic insight into both receptor flexibility and fragment interaction patterns. Importantly, this integrated strategy identified multiple non-purine scaffolds as promising chemical starting points for ligand optimization.

Spectral Shift technology played a central role by enabling fragment screening directly on active GPCR while maintaining native protein conformation, offering higher throughput and dramatically lower protein consumption than traditional biophysical approaches such as NMR, or SPR. In parallel, unbiased coarse-grained MD simulations captured long-timescale receptor dynamics and rapidly revealed potential fragment-binding hotspots, effectively narrowing down high-value regions of chemical space. These simulations also proved predictive, as suggested fragments were experimentally identified in Spectral Shift assays.

Together, this workflow provides an innovative, practical, and scalable solution for fragment-based GPCR drug discovery. Beyond accelerating early hit identification, it can serve as a prospective strategy to assess the likelihood of finding hits within focused or sub-libraries prior to initiating experimental campaigns, ultimately strengthening Integrated Drug Discovery efforts for complex membrane proteins.